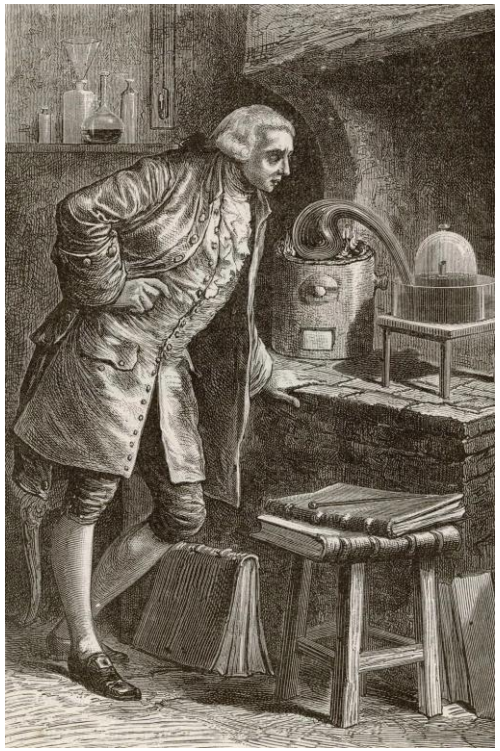
CY4001: Industrial Chemistry

Unit 3: Fuel Chemistry

Introduction

Antoine-Laurent Lavoisier (1743-1794)



Father of modern chemistry

- Proposed the law of conservation of mass
- Determined that combustion and respiration are caused by chemical reactions
- Discovered oxygen and its role in combustion and respiration.
- Systematized the naming of chemical compounds.

- Lavoisier came from a wealthy family
- He received a law degree and was admitted to the bar.
- He attended lectures in the natural sciences and created his own lab with his money

Social reformer

- Lavoisier was a humanitarian, he cared deeply about the people in his country and often concerned himself with improving the livelihood of the population by agriculture, industry, and the sciences.

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Course plan and organization

Unit 3: Fuel Chemistry

Review of energy sources (renewable and non-renewable). Classification of fuels (a) Coal: Classification - Pulverized Coal., Selection Criteria, Proximate and Ultimate Analysis, Liquefaction of Coal (Fischer-Tropsch method); Gasification of Coal (water gas). (b) Petroleum: Classification, Types of Cracking, Fuel Rating, LPG, Natural Gas, Producer Gas, Water Gas, Bio Gas. (c) Oxygenated fuels viz Biodiesel and Ethanol: their production and characterization

Learning Outcomes

- Understanding the nature and classification of fuels
- Chemistry behind the combustion process
- Identify the characteristics of Good Fuels
- Solid/Liquid/gas fuels and the interaction between the fuels

The earth has enough for every man's need but not for every man's greed.

Mahatma Gandhi

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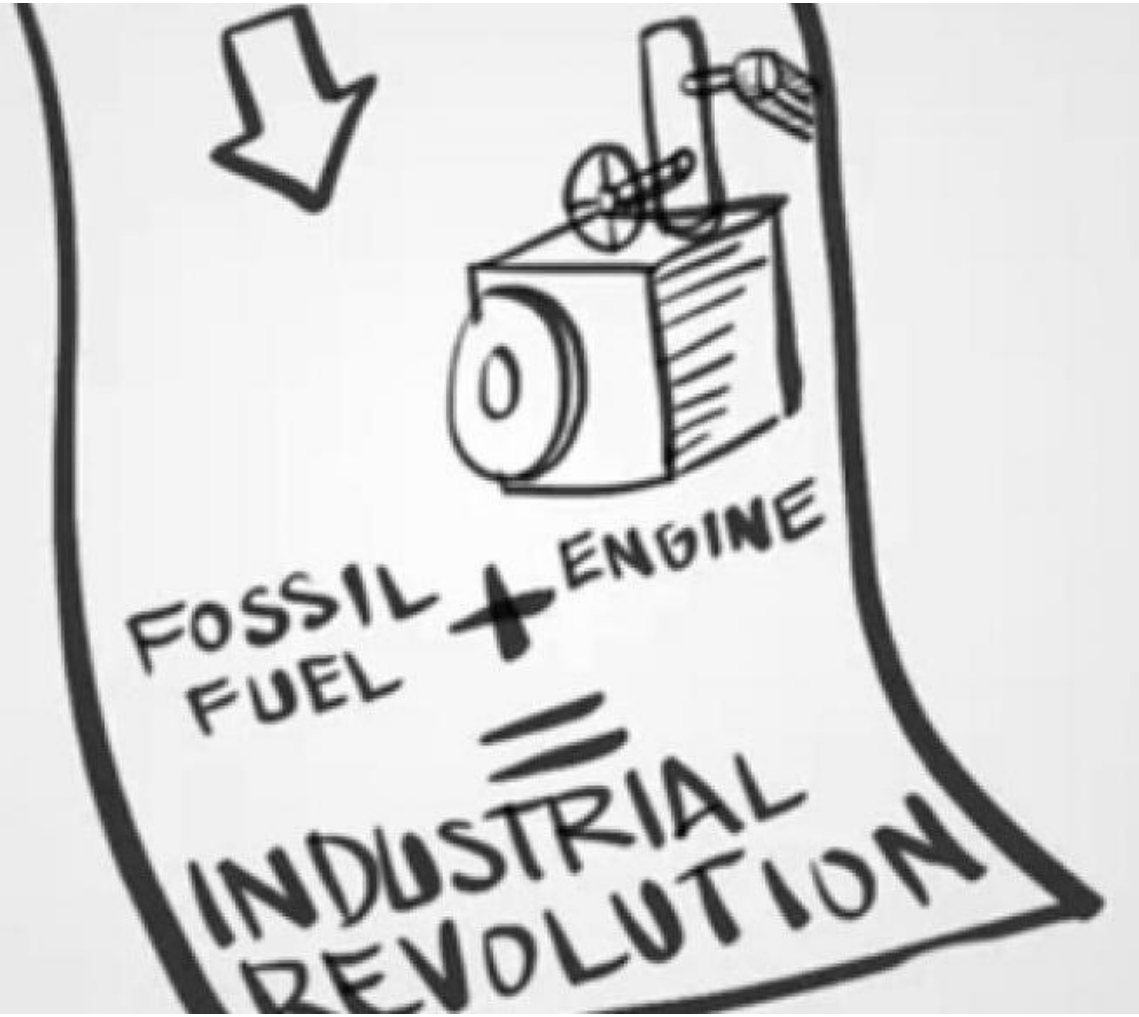
Why do we need to learn about fuels?



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Fuel:

Fuel is the source of heat energy, it can be stored as potential chemical energy and can be released through combustion

Combustible matter having carbon as a major ingredient, produce large amount of heat energy on burning and can be used for heat generation in industry and domestic applications is known as a fuel

Any compound or a substance which can produce energy and can be used in the production of power is termed as a fuel

In the combustion process, a fuel reacts with oxygen and releases the energy.

$$\text{Fuel} + O_2 \rightarrow CO_2 + H_2O + \text{heat}$$

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Introduction: Fuel Chemistry

- The increase of human population and living standards has resulted in a tremendous increase in the requirement of energy.

- Most of the developed and developing countries derive their energy from **fossil fuels** such as petroleum, coal and natural gas.

- The fossil fuels are nonrenewable - hence, they are depleting at an alarming rate.

- It is the basic responsibility of every citizen to conserve the nonrenewable resources.

Return-to renewables

Energy is a requirement in our everyday life as a way of improving human development leading to economic growth and productivity. The return-to renewables that helps to mitigate climate change is an excellent way but needs to be sustainable in order to ensure a sustainable future and bequeath future generations to meet their energy needs.

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Review of energy sources (renewable and non-renewable)

Non-Renewable Energy Sources

A nonrenewable resource is a substance that is used up more quickly than it can replace itself. **Example:** fossil fuels like oil, natural gas, and coal.

Renewable Energy Sources

Renewable energy sources replenish themselves naturally without being depleted.

Table 1. Renewable energy sources and their use (Panwar et al., 2011)

Energy sources	Energy conversion and usage options
Hydropower	Power generation
Morden biomass	Heat and power generation, pyrolysis, gasification, digestion
Geothermal	Urban heating, power generation, hydrothermal, hot dry rock
Solar	Solar home systems, solar dryers, solar cookers
Direct solar	Photovoltaic, thermal power generation, water heaters
Wind	Power generation, wind generators, windmills, water pump
Wave and tide	Numerous design, barrage, tidal stream

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Fuels and Combustion

Fuel is the source of heat energy; it can be stored as potential chemical energy and can be released through combustion.

In the combustion process, a fuel reacts with oxygen and releases the energy.

$$\text{Fuel} + O_2 \rightarrow CO_2 + H_2O + \text{heat}$$

Classification of fuels

- The fossil fuels are classified into the following categories:

  (i) Primary or natural fuels—These are found in nature.

  (ii) Secondary or artificial fuels—These are derived from primary fuels.

- Primary and secondary fuels may also be divided into three classes, namely solid, liquid and gaseous fuels.

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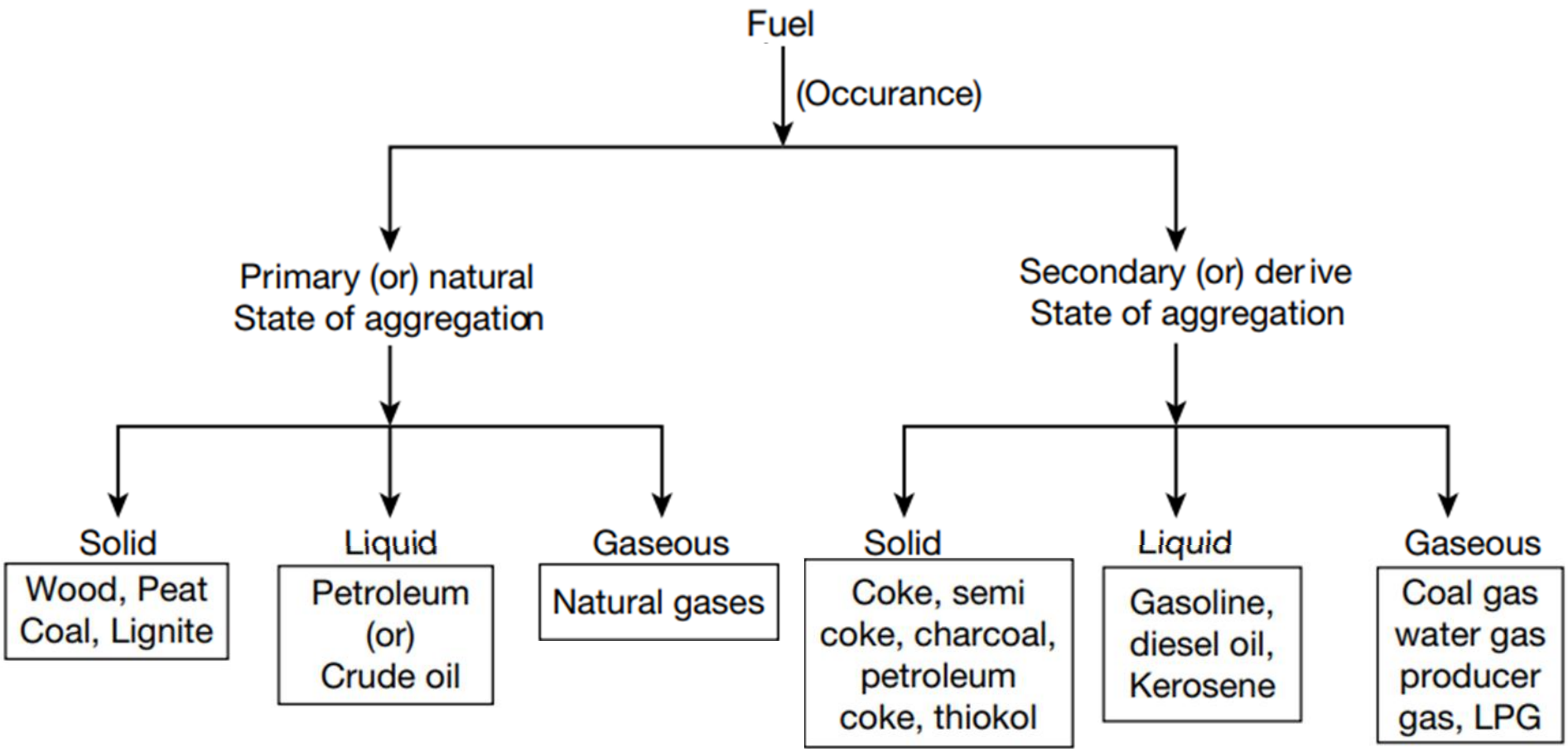
Classification of fuels



Peat in Ireland

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What are fossil fuels?

- Fossil fuels are hydrocarbons, primarily coal and petroleum (liquid petroleum or natural gas), formed from the fossilized remains of dead plants and animals.

- The term 'fossil fuel' also includes hydrocarbon-containing natural resources that are not derived entirely from biological sources, such as tar sands - properly known as "mineral fuels".

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Characteristics of Good Fuels

(a) High Calorific Value

- The amount of heat released is dependent on high calorific value, hence fuel should possess more HCV

(b) Moderate Ignition Temperature

- Low ignition temperature may result in a fire accident during storage and transportation.
- Very high ignition temperature may result in difficulties during the ignition.
- Hence, a good-quality fuel should have moderate ignition temperature.

(c) Low Quantity of Noncombustible Matters

- The presence of noncombustible matter will result in low calorific value, high quantity of ash generation and additional cost for transportation.
- Hence, a good fuel should have low level of noncombustible matter.

Characteristics of Good Fuels

(d) Lesser Moisture Content

The moisture present in the fuel will consume some amount of heat for its own evaporation and also reduce the calorific value, and increase the transportation and storage cost. The presence of moisture is not desirable in a good-quality fuel.

(e) Free from Harmful Gases like CO, SOx, H_2S

On combustion, the fuel should not release harmful and objectionable gases like CO, SOx, H_2S, NOx, *etc*.

(f) Moderate Velocity of Combustion

- Low rate of combustion may lead to unnecessary loss of heat due to radiation. A high rate of combustion leads to out of control burning.
- For a smooth and continuous supply of heat, the fuel must burn with moderate velocity.
- For easy start-up and stoppage, the combustion should be controllable.

Characteristics of Good Fuels

(g) Cost:

The fuel should have more availability at affordable cost. Storage cost of a fuel in bulk should be low

(h) Easy transportation:

Fuel should be easy to transport with low cost.

(i) Uniform particle size:

In case of solid fuels, for easy combustion, the particle size should be uniform

(j) High pyrometric effect:

Pyrometric effect is the highest temperature obtained with the fuel, hence fuel should have high pyrometric effect.

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Characteristics of Good Fuels

(i) *CGS system–calorie*: The amount of heat required to raise the temperature of 1 g of water through 1°C at 15 to 16°C.

(ii) *MKS system–kilo calorie*: The amount of heat required to raise the temperature of 1 kg of water through 1°C.

$$1 \text{ kcal} = 1{,}000 \text{ cal}$$

(iii) *British system (FPS system)*: *British thermal unit (BTU)*: The amount of heat required to raise the temperature of 1 pound of water through 1°F at 60 to 61°F.

$$1 \text{ BTU} = 252 \text{ cal} = 0.252 \text{ kcal}$$

$$1\text{kcal} = 3.968 \text{ BTU}$$

(iv) *Centigrade heat unit (CHU)*: The amount of heat required to raise the temperature of 1 pound of water through 1°C.

$$1 \text{ kcal} = 3.968 \text{ BTU} = 2.2 \text{ CHU}$$